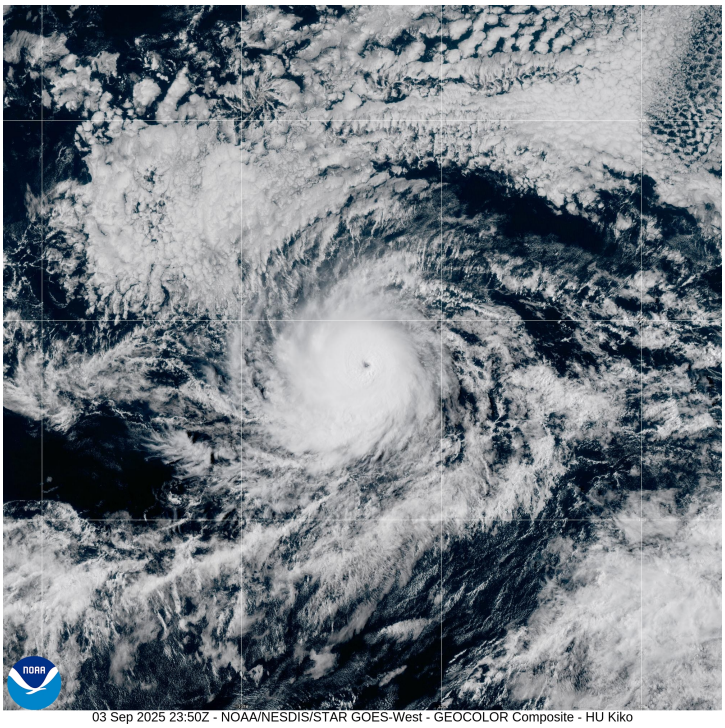
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

HURRICANE KIKO

(EP112025)

31 August – 10 September 2025

John L. Beven II
National Hurricane Center
20 November 2025



GOES-18 GEOCOLOR IMAGE OF KIKO NEAR PEAK INTENSITY AT 2350 UTC 3 SEPTEMBER. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Kiko was a category 4 hurricane (on the Saffir-Simpson Hurricane Wind Scale) over the western part of the eastern Pacific and the central Pacific basin. It later threatened the Hawaiian Islands, but passed north of the islands with little impact.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Kiko.

Hurricane Kiko

31 AUGUST – 10 SEPTEMBER 2025

BEST TRACK

The "best track[2]" positions and intensities for Hurricane Kiko are listed in Table 1. The best track chart of Kiko's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively. Observations also include flight-level, stepped frequency microwave radiometer (SFMR), and dropwindsonde observations from five flights of the 53rd Weather Reconnaissance Squadron of the U.S. Air Force Reserve Command.

The only observation of tropical-storm force winds during Kiko was from NOAA buoy 51000, which reported sustained winds of 37 kt at 1556 UTC 9 September and a peak gust of 41 kt as the center of Kiko passed to the southwest.

Origin

Kiko appears to have developed from a tropical wave that moved westward across Panama and Costa Rica on 22 August. This system moved westward and west-northwestward south of the coast of Mexico for the next several days with little development. Convection associated with the system first showed signs of organization on 29 August, and continued development led to the formation of a tropical depression at 0000 UTC 31 August about 825 n mi southwest of the southern tip of the Baja California Peninsula.

Peak Intensity and Minimum Pressure

Kiko's peak intensity of 125 kt early on 4 September is based on a combination of subjective and objective satellite intensity estimates. The minimum central pressure of 945 mb is based on the Knaff-Zehr-Courtney wind-pressure relationship for a storm of Kiko's size and 125-kt peak intensity.

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[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Kiko.

While Air Force Reserve Hurricane Hunter aircraft provided data on Kiko, they did not investigate the cyclone until several days after the peak intensity. The maximum 700-mb flight-level winds reported by the aircraft were 113 kt at 0601 UTC 7 September, and the minimum pressure reported by a dropwindsonde was 959 mb at 0604 UTC 7 September.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Kiko.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The genesis forecasts for Kiko were generally good, with the main deficiency being that the 2-day forecast genesis probabilities did not reach the high (>60% chance of development) category until 6 h before genesis. Another less skillful portion of the forecast occurred between 18-30 h before genesis, when the probabilities were briefly lowered as the pre-Kiko disturbance appeared to be developing more slowly than anticipated. Figure 4 shows composites of 7-day TWO genesis areas for each category prior to the formation of Kiko. Spatially, the genesis of Kiko was well forecast, with the genesis location being near the center of the genesis forecast areas.

A verification of official track forecasts for Kiko while it was located east of 140°W is given in Table 3a. Official track forecast errors were lower than the mean official errors for the previous 5-yr period at the forecast times from 12–96 h. The mean official errors were slightly above those for the previous 5-yr period at the 120-h forecast time. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. Many of the consensus models (entries TVCE through FSSE in Table 3b) have lower average errors than the official forecasts at most of the forecast times longer than 12 h. The best overall performance among the non-consensus models were from the experimental Google DeepMind AI model (GDMI) and the HWRF regional hurricane model (HWFI). Examination of the individual forecasts (not shown) suggests two forecast error issues: 1) The forecasts during the initial westward motion early in the cyclone's life were too far north, and 2) the forecasts during the west-northwestward motion in the middle part of the cyclone's life were too far south.

A verification of official intensity forecasts for Kiko while it was located east of 140°W is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period for the 24-72 h forecast times, and greater than the 5-yr means at the other times. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. The official forecast had lower mean errors than the vast majority of the intensity guidance, with only the HFIP Corrected Consensus model (HCCA) systematically having errors less than or close to the official forecasts. Examination of the individual forecasts (not shown) showed two significant error mechanisms. First, Kiko intensified more than forecast during

its rapid intensification to peak intensity. Second, near the end of the cyclone's life, Kiko weakened faster than forecast.

For the period when Kiko was west of 140°W, a verification of the official track forecasts for Kiko is given in Table 5a. A homogeneous comparison of the official track errors with selected guidance models is given in Table 5b. A verification of the official intensity forecasts for Kiko during this time is given in Table 6a, while a homogeneous comparison of the official intensity errors with selected guidance models is presented in Table 6b.

While Kiko passed within 60-70 n mi to the northeast of the western Hawaiian Islands, there were no coastal watches or warnings issued for the cyclone.

IMPACT-BASED DECISION SUPPORT SERVICES (IDSS) AND PUBLIC COMMUNICATION

The NHC began formal communication with emergency managers on Friday, September 5 as Kiko approached the Central Pacific basin. A decision support briefing was provided to emergency managers and coordinated through the FEMA Hurricane Liaison Team embedded at the NHC. The briefing was a federal video-teleconference with FEMA HQ and FEMA Region 9.

ACKNOWLEDGEMENTS

Philippe Papin (NHC) created the Graphical Tropical Weather Outlook verification figure, while Michael Spagnola (FEMA) provided the IDSS summary. The Central Pacific Hurricane Center and the Joint Typhoon Warning Center provided their fix information for Kiko.

Table 1. Best track for Hurricane Kiko, 31 August–10 September 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
31 / 0000	14.4	121.1	1007	30	tropical depression
31 / 0600	14.4	121.8	1007	30	"
31 / 1200	14.4	122.6	1006	35	tropical storm
31 / 1800	14.4	123.5	1006	35	"
01 / 0000	14.3	124.3	1004	40	"
01 / 0600	14.1	125.1	1004	40	"
01 / 1200	13.9	125.8	1002	45	"
01 / 1800	13.8	126.4	1001	55	"
02 / 0000	13.8	127.0	998	55	"
02 / 0600	13.8	127.6	993	65	hurricane
02 / 1200	13.8	128.1	987	75	"
02 / 1800	13.8	128.6	983	80	"
03 / 0000	13.8	129.2	974	90	"
03 / 0600	13.8	129.8	974	90	"
03 / 1200	13.9	130.4	974	90	"
03 / 1800	13.9	131.1	960	110	"
04 / 0000	13.8	131.9	945	125	"
04 / 0600	13.7	132.6	946	125	"
04 / 1200	13.8	133.3	952	115	"
04 / 1800	13.8	134.1	953	115	"
05 / 0000	13.8	135.0	956	110	"
05 / 0600	13.8	135.8	961	105	"
05 / 1200	14.0	136.6	962	100	"
05 / 1800	14.3	137.4	959	105	"
06 / 0000	14.7	138.2	952	115	"
06 / 0600	15.0	139.2	949	120	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
06 / 1200	15.3	140.1	953	110	"
06 / 1800	15.9	140.9	954	110	"
07 / 0000	16.3	141.9	951	115	"
07 / 0600	16.8	142.9	956	110	"
07 / 1200	17.2	143.9	962	105	"
07 / 1800	17.8	145.0	972	95	"
08 / 0000	18.5	146.0	974	95	"
08 / 0600	19.3	147.1	980	90	"
08 / 1200	20.0	148.2	981	85	"
08 / 1800	20.6	149.3	990	80	"
09 / 0000	21.3	150.3	995	65	"
09 / 0600	21.7	151.5	1004	50	tropical storm
09 / 1200	22.2	152.8	1006	40	"
09 / 1800	22.5	154.3	1007	40	"
10 / 0000	22.7	155.7	1007	35	"
10 / 0600	22.3	156.5	1007	35	"
10 / 1200	22.7	157.4	1008	30	remnant low
10 / 1800	23.4	158.4	1008	30	"
11 / 0000	24.3	159.5	1009	30	"
11 / 0600					dissipated
04 / 0000	13.8	131.9	945	125	maximum winds and minimum pressure

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	84	126
Medium (40%-60%)	66	90
High (>60%)	6	72

Table 3a. Official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Kiko, 31 August–10 September 2025 during the time it was east of 140°W. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**11.1**	**19.8**	**30.0**	**43.3**	**56.5**	**70.7**	**98.5**	141.2
OCD5	16.0	34.9	57.2	83.4	110.7	135.9	182.8	208.6
Forecasts	25	25	25	25	25	25	25	21
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Kiko, 31 August–10 September 2025 during the time it was east of 140°W. Errors smaller than the official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	10.7	18.4	27.2	39.7	50.7	61.0	87.6	133.3
OCD5	15.4	32.5	52.8	78.9	103.1	124.0	165.3	193.3
GFSI	13.2	20.8	28.9	40.9	**50.5**	63.1	101.3	144.6
HWFI	13.6	18.8	**20.7**	**28.5**	**32.1**	**39.0**	**71.7**	**116.2**
HMNI	13.7	20.4	27.3	**38.8**	51.9	62.7	95.7	138.5
HFAI	13.6	21.1	29.6	46.1	62.3	75.7	118.3	155.1
HFBI	15.2	22.9	29.9	42.3	54.8	69.5	118.5	180.9
UKMI	18.4	37.2	53.7	73.3	91.2	107.5	132.1	164.7
EMXI	15.0	23.1	34.1	50.9	69.8	85.5	108.7	141.9
CMCI	21.5	43.6	62.1	81.2	105.4	132.3	196.6	284.6
CTCI	17.9	23.8	29.3	**36.5**	**45.5**	**52.6**	95.7	181.2
TVCE	12.5	**18.2**	**24.3**	**33.8**	**39.4**	**45.1**	**69.6**	**107.2**
TVCX	12.2	**18.1**	**23.7**	**32.5**	**39.1**	**45.1**	**68.5**	**104.8**
GFEX	12.3	**17.3**	**23.0**	**32.5**	**43.7**	**53.4**	**78.5**	**114.6**
TVDG	12.3	**18.2**	**24.0**	**34.3**	**39.5**	**44.8**	**67.3**	**102.1**
HCCA	12.1	**17.5**	**26.3**	**38.8**	**49.4**	61.1	88.9	**132.4**
FSSE	11.3	**15.5**	**21.9**	**33.0**	**44.1**	**54.3**	**85.8**	133.7
AEMI	13.9	19.8	**25.9**	**36.5**	**49.4**	67.4	113.6	192.1
GDMI	12.4	18.7	**25.0**	**32.3**	**39.9**	**46.2**	**54.9**	**81.6**
TABS	36.0	85.3	131.8	171.5	200.9	217.2	206.9	222.1
TABM	16.3	32.3	49.8	69.1	86.3	105.2	147.0	160.9
TABD	12.0	**16.5**	**24.0**	**34.2**	53.0	85.0	166.9	204.7
Forecasts	21	21	21	21	21	21	21	18

Table 4a. Official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Kiko, 31 August–10 September 2025 during the time it was east of 140°W. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	7.2	**8.8**	**10.2**	**11.0**	**11.2**	**15.0**	19.8	21.4
OCD5	9.6	15.6	22.6	26.4	30.0	34.1	34.9	35.8
Forecasts	25	25	25	25	25	25	25	21
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Kiko, 31 August–10 September 2025 during the time it was east of 140°W. Errors smaller than the official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	7.0	9.3	10.9	11.4	11.8	15.9	19.3	21.8
OCD5	9.3	14.8	22.3	25.8	30.0	34.9	35.6	35.9
HWFI	11.1	13.9	17.0	16.6	18.9	19.9	19.7	**17.8**
HMNI	9.4	14.1	15.9	14.8	15.9	20.0	22.7	22.2
HFAI	8.5	15.3	17.0	15.9	15.4	17.1	**17.6**	**18.2**
HFBI	7.8	10.5	12.0	14.5	16.1	19.4	23.8	25.0
DSHP	8.7	13.0	19.2	24.2	27.3	29.1	27.3	26.1
LGEM	7.9	14.1	21.4	26.5	30.7	32.5	30.2	27.3
ICON	8.6	12.6	16.8	19.1	22.8	25.0	24.5	23.0
IVCN	7.7	12.2	16.5	18.4	21.6	24.1	23.9	22.3
IVDR	7.7	12.6	16.7	18.4	21.0	23.5	23.5	21.9
CTCI	8.6	15.0	22.2	27.4	32.3	37.3	39.9	39.2
GFSI	9.6	18.5	25.6	28.6	30.6	31.3	28.4	25.9
EMXI	11.0	18.8	25.5	29.9	33.4	36.1	35.9	35.1
HCCA	**6.7**	9.3	11.6	**11.0**	11.8	**14.7**	19.7	**21.1**
FSSE	8.0	10.9	14.8	15.0	16.9	19.6	22.4	27.1
GDMI	8.0	12.1	17.0	21.6	26.4	28.5	27.8	26.0
Forecasts	22	22	22	22	22	22	22	19

Table 5a. Official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Kiko, 31 August–10 September 2025 during the time it was west of 140°W. Mean errors for the previous eastern Pacific 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**17.0**	**26.3**	**33.8**	**35.6**	**43.6**	**59.0**		
OCD5	26.7	60.8	102.4	145.5	172.6	207.7		
Forecasts	14	12	10	8	6	4		
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4		
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0		

Table 5b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Kiko, 31 August–10 September 2025 during the time it was west of 140°W. Errors smaller than the official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 5a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	17.0	26.3	33.8	35.6	43.6	59.0		
OCD5	26.7	60.8	102.4	145.5	172.6	207.7		
GFSI	**16.9**	31.1	41.7	42.1	**42.6**	**47.9**		
HWFI	18.9	33.8	41.5	56.9	68.1	82.6		
HMNI	22.0	30.4	**30.0**	43.2	58.1	74.0		
HFAI	**16.8**	27.4	37.9	52.3	56.6	73.4		
HFBI	17.3	**24.5**	42.5	53.7	54.5	**58.2**		
UKMI	20.6	37.5	51.4	68.0	82.2	73.8		
EMXI	22.4	39.1	43.2	43.2	56.2	81.9		
CMCI	28.9	55.5	90.8	118.7	145.6	167.1		
CTCI	18.0	**25.7**	**31.8**	41.5	55.4	94.6		
TVCE	17.2	26.5	34.8	43.9	48.5	**58.5**		
TVCX	17.2	27.7	35.8	43.6	48.8	62.2		
GFEX	19.2	31.5	41.7	37.2	46.5	**58.2**		
TVDG	17.7	27.2	35.1	42.9	47.3	**55.2**		
HCCA	18.5	30.7	38.6	42.2	47.6	66.0		
FSSE	18.7	30.4	39.3	45.1	59.5	80.1		
AEMI	20.1	36.1	50.5	55.0	52.6	**29.2**		
GDMI	**16.5**	**20.1**	**28.1**	**31.9**	**36.9**	**41.4**		
TABS	26.7	56.3	87.2	104.6	108.1	97.2		
TABM	25.8	41.1	55.9	64.6	68.0	90.8		
TABD	46.3	103.6	155.1	215.4	279.8	358.5		
Forecasts	14	12	10	8	6	4		

Table 6a. Official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Kiko, 31 August–10 September 2025 during the time it was west of 140°W. Mean errors for the previous eastern Pacific 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.7	**5.4**	**8.5**	13.8	19.2	20.0		
OCD5	9.1	9.8	10.8	15.9	9.3	13.0		
Forecasts	14	12	10	8	6	4		
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5		
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3		

Table 6b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Kiko, 31 August–10 September 2025 during the time it was west of 140°W. Errors smaller than the official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 6a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.7	5.4	8.5	13.8	19.2	20.0		
OCD5	9.1	9.8	10.8	15.9	**9.3**	**13.0**		
HWFI	8.5	7.2	9.3	**12.0**	**12.2**	**12.5**		
HMNI	6.1	9.8	11.9	15.4	21.8	20.2		
HFAI	6.2	8.9	12.1	16.1	22.5	20.8		
HFBI	6.5	6.2	8.6	**10.6**	**14.3**	**13.0**		
DSHP	8.9	8.2	**8.1**	**7.5**	**6.5**	**10.0**		
LGEM	9.4	10.6	10.8	**9.4**	**6.8**	**6.5**		
ICON	7.4	7.7	8.5	**10.6**	**11.3**	**12.8**		
IVCN	7.1	6.8	**8.3**	**11.5**	**13.7**	**14.5**		
IVDR	6.7	7.2	8.8	**12.0**	**15.2**	**15.8**		
CTCI	8.4	7.7	**7.5**	**12.0**	**14.0**	**16.5**		
GFSI	7.3	8.9	9.6	**12.8**	**15.5**	**18.2**		
EMXI	6.9	15.3	21.6	27.2	35.2	39.2		
HCCA	6.5	**5.0**	**5.9**	**9.9**	**14.8**	**16.8**		
FSSE	6.9	5.8	**7.7**	**12.2**	**13.8**	**13.0**		
GDMI	7.7	7.3	**7.4**	**7.8**	**7.2**	**8.0**		
Forecasts	14	12	10	8	6	4		

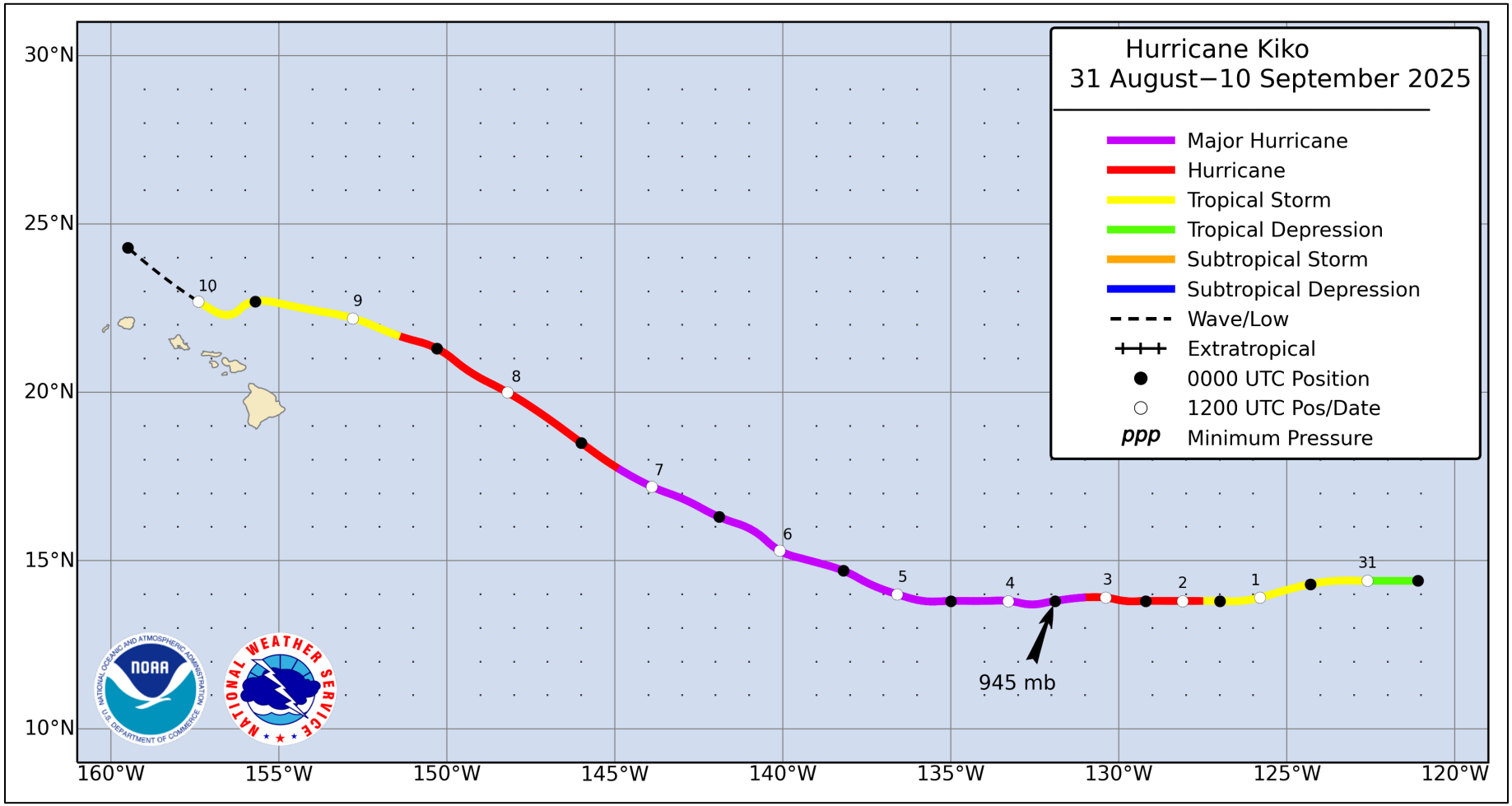


Figure 1. Best track positions for Hurricane Kiko, 31 August–10 September 2025.

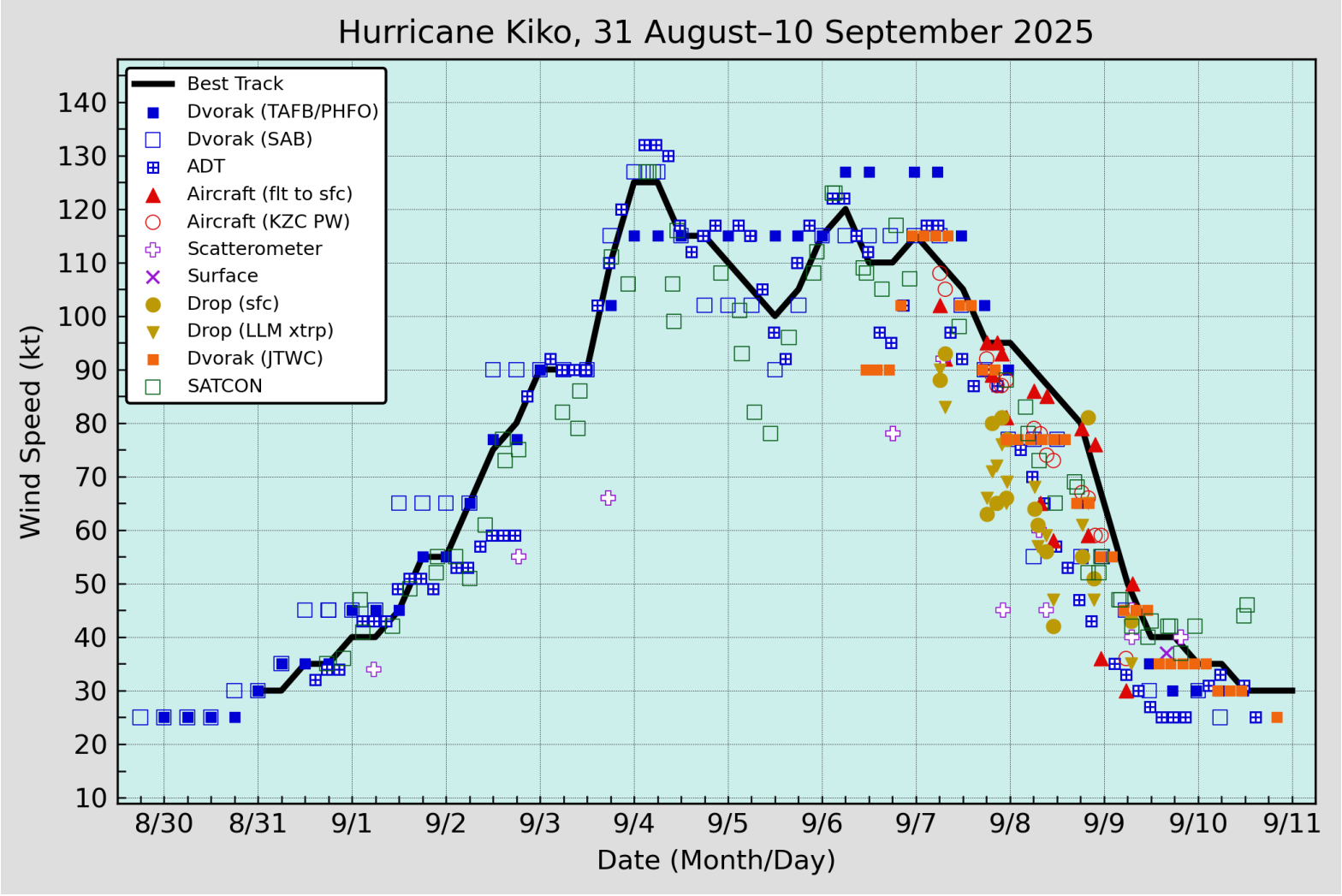


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Kiko, 31 August–10 September 2025. Aircraft observations have been adjusted for elevation using 90% and 80% adjustment factors for observations from 700 mb, and 850 mb, respectively. Dropwindsonde observations include actual 10 m winds (sfc), as well as surface estimates derived from the mean wind over the lowest 150 m of the wind sounding (LLM). Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

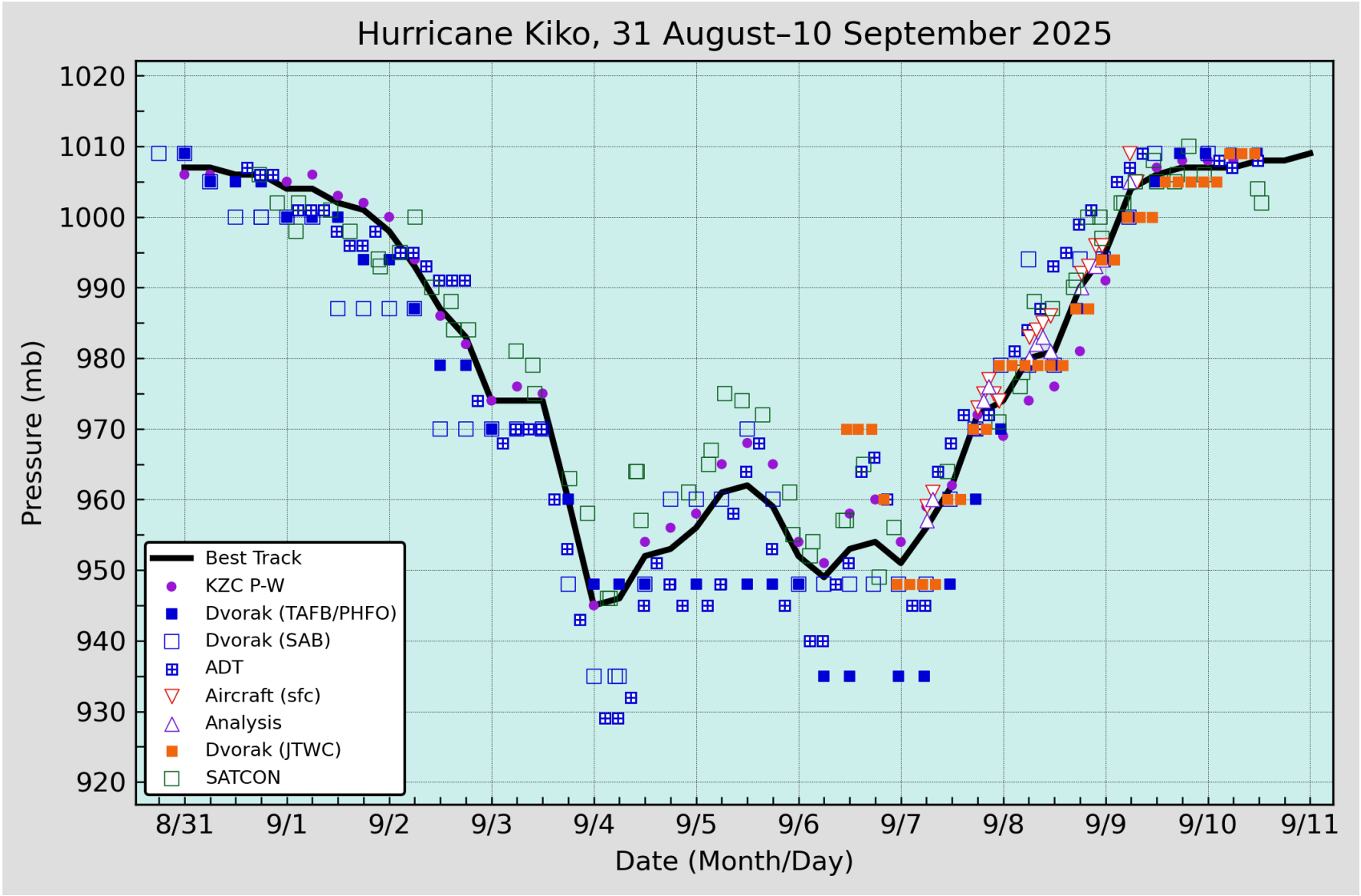


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Kiko, 31 August–10 September 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

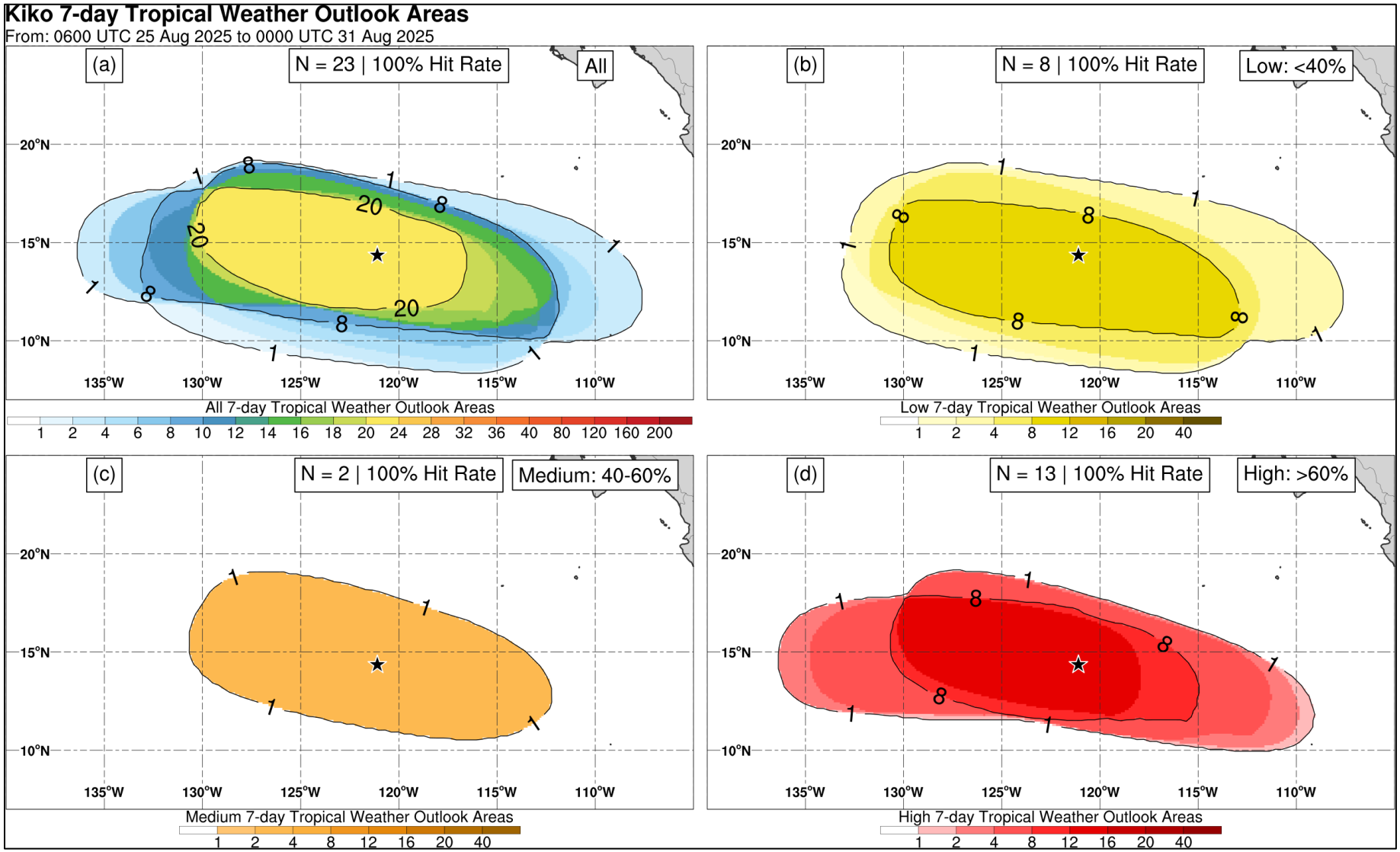


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Kiko for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.